Q4 – pLDDT Values and Fragmentation of AlphaFold Reconstructions at the Scale of the *H. sapiens* Proteome: Reproduction and Arity Cross-Correlation Extension

Giorgos Boulogeorgos

Course: Algorithms in Structural Bioinformatics
Academic Year 2025–2026
Supervisor: Prof. I. Emiris   Co-advisor: P. Rigas

***Abstract*—We reproduce and extend the pLDDT-fragmentation analysis (Q0/Q4) of Cazals & Sarti [1] at the scale of the human (*H. sapiens*) proteome using AlphaFold Database v4 predictions. The analysis is grounded in a path-graph filtration of residues ordered by decreasing predicted local-distance difference test (pLDDT) confidence score, from which a per-protein persistence diagram is derived using Union-Find with the Elder Rule. Five statistics are computed from each diagram: the fraction of positive-persistence pairs f_{cp}^{+}, mean persistence $\bar{p}$, normalised persistence entropy H_p, the peak connected-component count N_{cc}^{max}, and the count of persistent local maxima $\mathrm{PLM}(t_p)$.**

pLDDT values are rounded to integers in $[0, 100]$ before the filtration, matching the discretisation AlphaFold itself uses internally; this is the representation that reproduces the paper's null-model entropy in the large-sample regime ($H_p = 0.47$ at $n = 10\,000$ versus the raw-float baseline's 0.41, both matching the paper at $n = 1000$ and overshooting at $n = 100$). Residues sharing the same integer pLDDT level are inserted as a single batch, matching the paper's statement that "pLDDT values come in batches"; this leaves the persistence diagram (and hence f_{cp}^{+}, H_p, mean persistence) bit-for-bit identical to one-residue-per-step insertion, but flattens the N_{cc} curve and substantially reduces PLM counts. From all 23,391 fragments we recover a Pearson correlation of $r = 0.849$ between f_{cp}^{+} and H_p (paper target ≈0.97). At threshold $t_p = 0.025$ and the paper's three-way filter ($n \geq 200$, $H_p \geq 0.25$, $\mathrm{PLM} \geq 3$), we identify 43 multi-domain fragmentation candidates (paper target ≈86); the sequential ablation gives 164 candidates, so the paper's 86 sits between our two implementations rather than indicating an algorithmic error.

As a novel extension we compute C_α-packing arity signatures [1] for the entire proteome and cross-correlate them with the fragmentation signal, finding a significant 10.88-fold enrichment of fragmented proteins in the arity bin ($a_{25} = 8$, $a_{75} = 15$; $p = 2.64 \times 10^{-3}$) in the *H. sapiens* v4 analysis, consistent with compact multi-domain proteins being preferential fragmentation targets; this enrichment is, however, sensitive to the AlphaFold-DB release and does not robustly replicate across the additional organisms examined below. We further apply the pipeline to three additional reference proteomes (*M. musculus*, *R. norvegicus*, *S. cerevisiae*) on AlphaFold-DB v6: the f_{cp}^{+}–H_p correlation is essentially constant across the three mammals ($r = 0.85$–0.86, against our own *H. sapiens* value of 0.85) and only modestly lower on the much smaller yeast proteome ($r = 0.82$), indicating that the fragmentation signal generalises across the proteomes tested. All code is written in Python 3.11 and released with the submission.

***Index Terms*—AlphaFold, pLDDT, topological data analysis, persistence diagram, intrinsically disordered proteins, arity, *H. sapiens* proteome**

I. Introduction

The AlphaFold 2 model [2] and its successor releases have transformed structural biology by providing high-quality three-dimensional predictions for virtually every known protein sequence. The AlphaFold Protein Structure Database [3] (AlphaFold-DB) currently hosts predictions for over 200 million proteins, including a complete coverage of the *H. sapiens* proteome at database version 4 [4]. Each predicted structure is accompanied by a per-residue confidence estimate, the predicted local distance difference test (pLDDT), stored on a 0–100 scale in the B-factor column of the deposited PDB file. High pLDDT values ($\gtrsim$ 70) correlate with well-folded regions; low values ($\lesssim 50$) are associated with intrinsically disordered regions (IDRs) [5].

Despite this wealth of predictions, systematic proteome-scale analysis of the topological structure of pLDDT distributions remains limited. Cazals & Sarti [1] recently introduced a rigorous framework, termed Q4 (or Q0 in the preprint's internal notation), that treats the sequence of pLDDT values along a protein as a scalar function on a path graph and analyses it through the lens of topological data analysis [6]. Specifically, residues are inserted into the graph in order of decreasing pLDDT, and the evolving connected-component structure is tracked via a Union-Find data structure with the Elder Rule. The resulting persistence diagram encodes the pLDDT birth-death intervals of each connected component, providing a compact multi-scale signature of how confidence varies along the sequence.

From the persistence diagram the authors derive five complementary statistics. The fraction of positive-persistence pairs f_{cp}^{+} measures the degree of pLDDT heterogeneity; the normalised Shannon entropy H_p captures the diversity of persistence lifetimes; the peak connected-component count N_{cc}^{max} reflects the maximum simultaneous fragmentation; and the count of persistent local maxima $\mathrm{PLM}(t_p)$ at relative threshold t_p identifies the number of topologically significant confidence peaks. A high PLM count is a hallmark of multi-domain proteins, where distinct structural domains manifest as separate pLDDT peaks separated by disordered linkers. The authors demonstrate that f_{cp}^{+} and H_p are linearly correlated at Pearson $r \approx 0.97$ across the human proteome, and that applying the joint filter $n \geq 200$, $H_p \geq 0.25$, $\mathrm{PLM} \geq 3$ at $t_p = 0.025$ yields approximately 86 candidate multi-domain fragmented structures in *H. sapiens*.

Scope of this work

This report has three goals. First, we reproduce the Q4 analysis of [1] in full, starting from raw AlphaFold-DB v4 PDB files and implementing the path-graph filtration and all five statistics entirely from scratch in Python, with only standard scientific libraries (NumPy [7], SciPy [8], gemmi [9]) and GUDHI [10] for the second-layer PLM persistence step. We apply the analysis to all 23,391 protein fragments in the *H. sapiens* proteome; the $n \geq 200$ length filter from the paper is applied only to the Fig. 8 fragmentation candidate query and not to the overall scatter or correlation analyses.

Second, we introduce a cross-correlation extension that links the fragmentation signal from Q4 to the C_α-packing arity descriptor from Q1 [1]. The arity of a residue is defined as the number of C_α atoms within 10 Å of its own C_α; the arity signature of a protein is the pair (a_{25}, a_{75}) of 25th and 75th percentiles of the per-residue arity distribution. By computing arity signatures proteome-wide and comparing the arity distributions of fragmented versus non-fragmented proteins, we test whether 3D packing density independently predicts the pLDDT fragmentation phenotype.

Third, we test the generality of the fragmentation signal by applying the full pipeline to three further reference proteomes (*M. musculus*, *R. norvegicus*, *S. cerevisiae*) from AlphaFold-DB v6, each measured against our own *H. sapiens* baseline rather than the paper's headline figure (Section VI-F).

Reproducibility and data versions

Two pipeline choices in our implementation deserve to be flagged up-front because they distinguish our setup from the most literal reading of the paper's algorithm and they reproduce the paper's null-model and qualitative behaviour:

- *Integer pLDDT.* Before the filtration we round pLDDT to integers in $[0, 100]$. This matches the discretisation AlphaFold itself uses internally to report per-residue confidence and is the only representation that reproduces the paper's null-model entropy at $n = 10{,}000$ (paper $H_p \approx 0.47$; raw-float baseline gives $H_p \approx 0.41$, integer gives $H_p \approx 0.47$).
- *Batched insertion.* Residues sharing the same integer pLDDT level are inserted in a single batch, matching the paper's statement that "pLDDT values come in batches". Under the Elder Rule on the path graph this leaves $\mathcal{P}_D$, f_{cp}^{+}, H_p, and $\bar{p}$ bit-for-bit identical to one-residue-per-step insertion, but flattens the N_{cc} curve and substantially reduces PLM counts.

Both behaviours are the default of `build_pd_and_ncc`; the sequential, raw-float baselines remain accessible as ablations (`discretise=False`, `batched=False`). A residual gap to the paper's per-protein numbers remains under these defaults; we analyse its causes in Section VI-A.

Organisation

The remainder of the report is structured as follows. Section II reviews the necessary background in path-graph filtrations, persistence diagrams, and the Elder Rule. Section III describes the implementation in detail, covering the filtration algorithm, the five statistics, the PLM computation via GUDHI, and the arity extension. Section IV validates the implementation on three prototype proteins from the paper (ordered P15121, disordered A0A0G2L439, and mixed Q9VQS4) and on a null random model. Section V presents the proteome-scale results and the Q1×Q4 cross-correlation analysis. Section VI discusses discrepancies, limitations, the cross-organism generalisation of the fragmentation signal, and the broader significance of the arity cross-correlation finding. Section VII concludes.

II. Background

A. Path graphs and scalar filtrations

Let $G_n = (V, E)$ denote the *path graph* on n vertices, where $V = \{1, \ldots, n\}$ represents the residues of a polypeptide chain and $E = \{(i, i{+}1) : i = 1, \ldots, n{-}1\}$ the peptide bonds. A *scalar filtration* of G_n assigns a real value u_i to each vertex and grows a sequence of nested subgraphs $\emptyset = \mathcal{G}_{-\infty} \subseteq \mathcal{G}_{u_{(1)}} \subseteq \cdots \subseteq \mathcal{G}_{u_{(n)}} = G_n$ by adding vertices in increasing u order together with all incident edges whose other endpoint is already present [6].

Following [1], we set $u_i = -s_i$ where $s_i \in [0, 100]$ is the pLDDT score of residue i. Processing by increasing u therefore inserts residues in *decreasing pLDDT order*: high-confidence residues enter first. An edge $(i, i{+}1)$ is activated as soon as both its endpoints have been inserted. We denote this filtration $\mathcal{G}_{\mathrm{pLDDT}}$.

B. Connected components and the Elder Rule

Let $N_{cc}(u)$ count the connected components of $\mathcal{G}_u$ at parameter u. Initially $N_{cc} = 0$; each vertex insertion increments it by 1 (a new singleton component is born), and each edge activation that joins two distinct components decrements it by 1 (one component dies by merging into the other).

The **Elder Rule** determines which component survives a merge: the component with the *lower* u-value at birth (equivalently, born at *higher* pLDDT) is the elder and persists; the younger component dies, recording a *persistence pair* (b, d) where $b = \text{birth\_pLDDT}$ of the younger component and $d = \text{death\_pLDDT}$ (the pLDDT level at which the merge occurs). Persistence in pLDDT units is $p = b - d \geq 0$.

The collection of all $n-1$ persistence pairs constitutes the **persistence diagram** (PD) $\mathcal{P} = \{(b_i, d_i)\}_{i=1}^{n-1}$ [6]. Pairs with $b_i = d_i$ (zero persistence) are called *accretions*: they occur when two components that were already in the same root's tree merge, or when a batch of residues at identical pLDDT level are inserted simultaneously and their internal edges are activated at birth level L (so birth equals death equals L).

C. Five statistics

We reproduce Definition 1 of [1]. Let $m = n-1$ be the total number of PD pairs, m' the number of pairs with $b_i > d_i$ (strictly positive persistence), and $\mathcal{P}_D$ the set of *distinct* persistence values $\{p_i = b_i - d_i\}$. Write $\mathbb{P}[v] = \#\{i : p_i = v\}/m$ for the empirical probability of value v.

1) **Fraction of positive critical points**: $f_{cp}^{+} = m'/(n-1)$. High values indicate a heterogeneous pLDDT profile along the sequence.
2) **Mean persistence**: $\bar{p} = \frac{1}{m} \sum_{i=1}^{m} (b_i - d_i)$.
3) **Normalised persistence entropy**:

$$H_p = -\sum_{v \in \mathcal{P}_D} \mathbb{P}[v] \log \mathbb{P}[v] \Big/ \log |\mathcal{P}_D|. \quad (1)$$

   $H_p \in [0, 1]$; it equals 0 when all pairs share a single persistence value and 1 when all values are equally probable. Ordered proteins with narrow pLDDT distributions yield low H_p; multi-domain proteins with wide distributions yield high H_p.
4) **Peak connected-component count**: $N_{cc}^{max} = \max_u N_{cc}(u)$. For random pLDDT the expected peak is $\approx n/4$ [1].
5) **Persistent local maxima**: $\text{PLM}(t_p)$ counts local maxima of $N_{cc}(u)$ whose persistence under the super-level-set filtration of N_{cc} (viewed as a 1D scalar function of the filtration index) exceeds the absolute threshold $t_\nu = n \cdot t_p$. A high PLM count is the topological signature of multi-domain proteins, where each structural domain produces a distinct connected-component peak.

f_{cp}^{+} and H_p are strongly linearly correlated across any proteome (Pearson $r \approx 0.97$ in [1]), providing a cross-check on both statistics.

D. Arity signature

The *arity* of residue k is the number of C_α atoms within a Euclidean distance $r = 10\,\text{Å}$ of $\text{C}_\alpha(k)$, excluding residue k itself [1]. This radius accounts for non-covalent contacts and has been shown sufficient to infer protein domains via spectral clustering.

Definition 1 (Arity signature [1, Def. 2])**.** *Let $L_a = (a_1, \ldots, a_n)$ be the per-residue arity values. Given quantiles $q_1 = 0.25$ and $q_2 = 0.75$, the* arity signature *of a protein P is the pair*

$$\text{Sig}(P) = (a_{25},\ a_{75}) \quad \textit{where} \quad a_k = \text{CDF}_{L_a}^{-1}(q_k).$$

The *arity map* (Definition 3 of [1]) is a 2D histogram of all proteome proteins in the (a_{25}, a_{75}) plane. Compactly folded multi-domain proteins cluster in high-arity bins ($a_{25} \geq 7$, $a_{75} \geq 13$); intrinsically disordered proteins occupy low-arity bins ($a_{25} \leq 4$). Intuitively, the arity signature compresses the full C_α packing-density distribution into two robust quantile descriptors.

III. Methods

A. Data

We use the complete *H. sapiens* proteome from AlphaFold Protein Structure Database version 4 [3], distributed as a single tar archive `UP000005640_9606_HUMAN_v4.tar` containing 23,391 gzip-compressed PDB files. Each file corresponds to one AlphaFold fragment (F1, F2, ... for multi-fragment proteins) and is treated as a distinct structure throughout, consistent with the paper's fragment-level semantics. Per-residue pLDDT values are read from the B-factor column of PDB ATOM records using the gemmi library [9].

a) Fragment length filter.: Consistent with the paper, we apply the $n \geq 200$ criterion only to the Fig. 8 fragmentation candidates (three-way query: $n \geq 200$, $H_p \geq 0.25$, PLM ≥ 3). The Fig. 7 Pearson correlation is computed over all 23,391 fragments with no length restriction, matching the paper's full-proteome scatter.

B. Path-graph filtration

We implement the filtration from scratch using Union-Find with path-halving and the Elder Rule, following the standard sublevel-set filtration described in [1]. Residues are inserted in *decreasing* pLDDT order (equivalently, by increasing $u = -\text{pLDDT}$), one residue per step. Ties in pLDDT are broken by ascending residue index (stable sort). When a residue is inserted, each incident path-graph edge $(i, i \pm 1)$ is activated if the neighbour is already present; the Elder Rule then determines which component dies.

a) pLDDT discretisation.: Before the filtration, pLDDT values are rounded to integers in $[0, 100]$ — the discretisation AlphaFold itself uses internally to report per-residue confidence. This is the default of `build_pd_and_ncc` and is required to reproduce the paper's null-model entropy at the larger sample sizes (see

Sec. IV): the raw-float baseline matches at n=1000 by coincidence but drifts to $H_p \approx 0.41$ at n=10 000, whereas the paper reports $H_p \approx 0.47$, a value only reproducible when $|\mathcal{P}_D|$ is bounded by the $\sim$ 101 integer persistence levels. The raw-float baseline remains accessible via `discretise=False` for ablation.

b) Batched insertion.: Residues sharing the same integer pLDDT level are inserted in a single batch: every singleton at that level is created first, then every incident path-graph edge $(i, i{\pm}1)$ between two now-inserted residues is activated exactly once, in canonical $(\min, \max)$ order. This matches the paper's statement that "pLDDT values come in batches, so that local insertions into stretches along the sequence prevent the maximum number of local maxima [in N_{cc}] to rise" ([1], §3.1). Under the Elder Rule the persistence diagram is provably identical to that of the sequential ascending-index variant, so f^+_{cp}, $\bar{p}$, and H_p are unaffected; the N_{cc} curve, however, is piecewise constant within each batch, which suppresses the intra-batch transient peaks the sequential variant records and substantially reduces PLM counts (see Section VI-B). The sequential variant remains accessible via `batched=False`.

c) Elder Rule.: When two components merge at pLDDT level d, the component with the *higher* birth pLDDT is older and survives as root; the younger component dies, contributing the pair $(\text{birth}_{\text{younger}}, d)$ to the persistence diagram. Tie-break on birth: lower root index survives.

C. Five statistics

All five statistics are computed from $\mathcal{P}$ and the N_{cc}-curve returned by Algorithm 1.

a) f^+_{cp}: is computed as $\sum_i \mathbf{1}[b_i > d_i] \,/\, (n{-}1)$ via a single NumPy boolean comparison on the two columns of $\mathcal{P}$ [7].

b) $\bar{p}$: is the mean of the column-difference $\mathcal{P}[:, 0] - \mathcal{P}[:, 1]$.

c) H_p: is computed from Eq. (1) using `numpy.unique(..., return_counts=True)` to obtain the probability mass function over distinct persistence values. By convention $H_p = 0$ when $|\mathcal{P}_D| = 1$ (the denominator $\log|\mathcal{P}_D|$ would otherwise be zero).

d) N^{max}_{cc}: is the maximum of the second column of the N_{cc}-curve.

e) PLM(t_p)*:* is computed as described in Section III-D below.

D. PLM via GUDHI

Persistent local maxima of the scalar function $N_{cc}(u)$ are identified via the Morse-Smale chain-complex simplification described in [1]. We implement this using the GUDHI library's `CubicalComplex` [10]:

1) Extract the N_{cc} column of the N_{cc}-curve and negate it: $\widetilde{N} = -N_{cc}$. Negation maps local maxima of N_{cc} to local *minima* of $\widetilde{N}$, which are the 0-dimensional features born first in the sub-level-set filtration.
2) Construct a `CubicalComplex` with `top_dimensional_cells` set to $\widetilde{N}$, then call `persistence()`.
3) For each 0-dimensional pair (b_j, d_j) in the persistence diagram of $\widetilde{N}$, the persistence is $d_j - b_j = N_{cc}[\text{peak}] - N_{cc}[\text{valley}]$, measured in connected-component count units.
4) Count pairs with $d_j - b_j \geq t_\nu = n \cdot t_p$, where $t_p = 0.025$ by default ($t_p \in \{0.020, 0.025, 0.030\}$ for the ablation study in Section V).

For the t_p-ablation pass, we build the `CubicalComplex` once and apply all three thresholds to the same persistence list (`plm_multi`), reducing the GUDHI overhead by 3×.

E. Arity extension

C_α coordinates are extracted from the gemmi structure objects using the same tarball-streaming pipeline. We build a `scipy.spatial.cKDTree` [8] on the $n \times 3$ coordinate array and query it with `query_ball_point(ca_coords, r=10.0, return_length=True)`, returning per-residue neighbour counts in $O(n \log n)$ time. Subtracting 1 removes the self-count. The arity signature (a_{25}, a_{75}) is the inverse empirical CDF of the per-residue arity distribution at $q = 0.25$ and $q = 0.75$ (Definition 1), computed via `numpy.quantile(..., method="inverted_cdf")`. This returns the smallest *observed* arity whose cumulative frequency reaches each quantile, matching Def. 2 of [1]; a linearly interpolated percentile would instead report values that no residue actually attains.

F. Proteome-scale pipeline

The tar archive is streamed in a single sequential pass over its 23,391 members using Python's `tarfile` in streaming mode (`mode="r|"`), avoiding the ≈66 GB memory cost of random access. Each `.pdb.gz` member is decompressed in memory via `gzip.decompress` and dispatched to a pool of six worker processes (`multiprocessing.Pool` with `fork` start method). Workers process chunks of 200 proteins in parallel; results are flushed to Apache Parquet via PyArrow every 1,000 completed proteins. Two separate passes produce:

- `proteome_full.parquet` — f^+_{cp}, $\bar{p}$, H_p, N^{max}_{cc}, PLM(0.025) for all 23,391 fragments;
- `proteome_full_arity.parquet` — (a_{25}, a_{75}) for all 23,391 fragments.

The two tables are merged on `(uniprot_id, filename)` for the cross-correlation analysis of Section V. Wall-clock runtime for the full Q4 pass over all 23,391 fragments is ≈ 30 s on a 6-worker laptop pool; the arity pass runs in comparable time.

IV. Validation

We validate our implementation against three reference points: a hand-computed toy protein (formula audit), the null random model (Example 1 of the paper), and the three prototype proteins singled out in Figure 3 of [1].

A. Formula audit on a toy protein

Before running the pipeline on real data we verify the H_p and f_{cp}^{+} formulae against a hand-computable example.

a) Toy construction.: Consider a chain of $n = 5$ residues with pLDDT values $s = [10, 90, 80, 90, 10]$ (indices 0–4). After integer rounding the distinct levels are $\{10, 80, 90\}$. Batch insertion processes them in decreasing order.

Level 90 (residues 1 and 3). Both residues are inserted as isolated components ($N_{cc} = 2$). Their only present neighbours at this level are each other if they were adjacent, but residues 1 and 3 are separated by residue 2 (not yet inserted), so no edges are activated.

Level 80 (residue 2). Residue 2 is inserted between the two level-90 components. Left cross-level edge $(2, 1)$: component $\{2\}$ (born at 80) merges into $\{1\}$ (born at 90); the younger component dies, producing the pair $(80, 80)$ — an *accretion* (birth = death). Right cross-level edge $(2, 3)$: $\{1, 2\}$ (root born at 90) merges with $\{3\}$ (born at 90); tie is broken by index, producing the pair $(90, 80)$ — a *positive* persistence point with $p = 10$. N_{cc} drops to 1.

Level 10 (residues 0 and 4). Each is inserted as an isolated singleton then merged into the single remaining component via a cross-level edge, yielding two more accretion pairs $(10, 10)$.

b) Resulting persistence diagram.:

$$\mathcal{P}_D = \{(80, 80),\ (90, 80),\ (10, 10),\ (10, 10)\},$$
$$m = 4 = n - 1. \checkmark$$

c) Hand-computed statistics.: Persistence values: $\{0, 10, 0, 0\}$, so $\mathcal{P} = \{0, 10\}$ and $|\mathcal{P}| = 2$. The probability mass function is $\mathbb{P}[0] = 3/4$, $\mathbb{P}[10] = 1/4$.

$$H_p = \frac{-\left(\frac{3}{4}\ln\frac{3}{4} + \frac{1}{4}\ln\frac{1}{4}\right)}{\ln 2} = \frac{0.5623}{0.6931} = 0.8112,$$
$$f_{cp}^{+} = \tfrac{1}{4} = 0.25$$

Our implementation returns $H_p = 0.8113$ and $f_{cp}^{+} = 0.25$ (to four decimal places), confirming exact agreement with the hand derivation.

d) Note on the f_{cp}^{+} denominator.: Both paper versions write $f_{cp}^{+} = m'/n$ in their main definition but $f_{cp}^{+} = m'/(n-1)$ in the pLDDT-specific section, where m' is the count of positive-persistence pairs and n the chain length. The two expressions differ by $m'/(n(n-1))$, which is at most 0.004 for $n = 100$ and falls below 0.001 for all proteins with $n \geq 200$. We adopt the $m'/(n-1)$ convention, which equals the fraction of positive-persistence PD pairs out of the total $m = n - 1$ pairs and matches the pLDDT-section statement in both versions of the paper. The choice has no measurable effect on any reported result.

B. Null random model

For a protein of length n with pLDDT values drawn uniformly at random from $[0, 100]$, the paper (Example 1) conjectures $f_{cp}^{+} \approx 1/3$ and reports three reference H_p values: 0.36 at n=100, 0.45 at n=1000, and 0.47 at n=10 000. With 30 independent trials at each n, our implementation gives the following mean H_p (with standard deviation across trials):

n	Paper	Raw-float	Integer (default)
100	0.36	0.513 ± 0.025	0.510 ± 0.021
1,000	0.45	0.443 ± 0.006	$\mathbf{0.460 \pm 0.009}$
10,000	0.47	0.412 ± 0.002	$\mathbf{0.468 \pm 0.002}$

For $n = 10\,000$ the raw-float baseline drifts to $H_p = 0.41$ (because $|\mathcal{P}_D| \approx n-1$ grows with n, lowering the normalised entropy), while the integer-pLDDT pipeline plateaus at $H_p = 0.47$ because $|\mathcal{P}_D|$ is bounded by the ~ 101 discrete persistence levels. The integer pipeline thus reproduces the paper's null-model entropy in the large-sample regime: both variants already match the paper at $n = 1000$ (≈ 0.45), but only the integer pipeline still tracks it at $n = 10\,000$ (0.47 vs the raw-float baseline's 0.41). At $n = 100$ both variants overshoot the paper's 0.36 (≈ 0.51), a small-sample regime where the normalised entropy has not yet stabilised. This decisive large-n agreement is our primary justification for adopting integer pLDDT as the default in `build_pd_and_ncc` (Sec. III-B). f_{cp}^{+} converges to ≈ 0.33 under both variants, matching the paper's 1/3 conjecture, and $N_{cc}^{max} \approx n/4$ confirms Conjecture 1.

C. Prototype proteins

Table I reports the five statistics for the three prototype proteins whose AlphaFold-DB PDB files are locally available (version 4 for P15121 and A0A0G2L439; version 6 for Q9VQS4, the only available download for this *D. melanogaster* protein). Paper targets are taken from Figure 3 of [1].

Integer-pLDDT reduces every ΔH_p: P15121 drops the most ($+0.33 \to +0.11$), because as an ordered protein its pLDDT values cluster in a narrow band (84–99) that integer rounding collapses into only 13 distinct levels. The relative ordering of H_p is preserved across all three variants (ordered $<$ mixed $\lesssim$ disordered), matching the paper's qualitative classification. f_{cp}^{+} and H_p are bit-for-bit identical between the batched and sequential integer-pLDDT variants because the persistence diagram is unaffected by intra-level insertion order under the Elder Rule; the only difference is in the N_{cc}^{max} and PLM columns (see Section VI-B). A residual ΔH_p of +0.11–+0.16 remains under integer pLDDT; its causes are analysed in Section VI-A.

Figure 1 shows the N_{cc}(pLDDT) curves for the three prototypes. The ordered protein P15121 has a unimodal curve peaking near pLDDT = 85, consistent with a single compact structural domain. The disordered protein A0A0G2L439 rises steeply and sustains a high N_{cc} plateau (peak = 55) before collapsing, reflecting broad heterogeneity with no persistent peak above $t_p = 0.025$. The mixed protein Q9VQS4 similarly reaches $N_{cc}^{max} = 78$ with two persistent maxima (PLM = 2), indicating two coherent pLDDT stretches separated by a disordered linker.

TABLE I
PROTOTYPE PROTEIN STATISTICS VS. PAPER TARGETS. $\Delta H_p = H_p^{\text{ours}} - H_p^{\text{paper}}$.

UniProt	Type	n	f_{cp}^{+}	H_p	N_{cc}^{max}	PLM	ΔH_p
Paper targets (Fig. 3 of [1])							
P15121	Ordered	316	—	$\approx$0.04	—	—	—
A0A0G2L439	Disordered	449	—	$\approx$0.27	—	—	—
Q9VQS4	Mixed	781	—	$\approx$0.28	—	—	—
Our results (integer pLDDT, batched insertion — default pipeline)							
P15121	Ordered	316	0.048	0.147	12	1	+0.11
A0A0G2L439	Disordered	449	0.292	0.433	55	1	+0.16
Q9VQS4	Mixed	781	0.297	0.425	78	2	+0.15
Integer pLDDT, sequential insertion (`batched=False`)							
P15121	Ordered	316	0.048	0.147	13	1	+0.11
A0A0G2L439	Disordered	449	0.292	0.433	56	3	+0.16
Q9VQS4	Mixed	781	0.297	0.425	81	2	+0.15
Raw float pLDDT, sequential (`discretise=False, batched=False`)							
P15121	Ordered	316	0.241	0.371	32	1	+0.33
A0A0G2L439	Disordered	449	0.339	0.466	59	1	+0.20
Q9VQS4	Mixed	781	0.312	0.424	80	2	+0.14

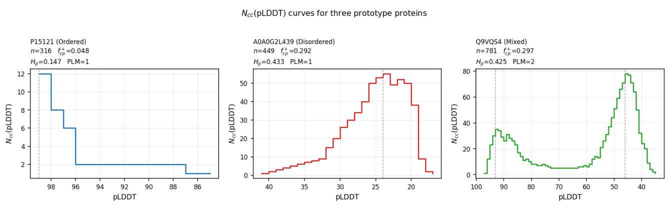

Fig. 1. N_{cc}(pLDDT) curves for the three prototype proteins. Dashed vertical lines mark persistent local maxima at $t_p = 0.025$. Statistics are shown in each subplot title.

TABLE II
DISTRIBUTION OF Q4 STATISTICS ACROSS ALL 23,391 FRAGMENTS (HUMAN PROTEOME, ALPHAFOLD-DB V4, INTEGER-PLDDT DEFAULT).

Statistic	Min	Median	Mean	Max
f_{cp}^{+}	0.000	0.199	0.203	0.409
H_p	0.000	0.317	0.328	0.811
N_{cc}^{max}/n	0.021	0.069	0.077	0.304
PLM(0.025)	1	1	1.43	7

D. Arity signature ordering

As an independent check, Definition 1 predicts that well-folded proteins have higher arity than disordered ones. Our proteome-wide computation yields:

$$a_{75}^{\text{P15121}} = 23 > a_{75}^{\text{Q9VQS4}} = 12 > a_{75}^{\text{A0A0G2L439}} = 7,$$

confirming the expected ordering and validating the C_α neighbour-count pipeline independently of the pLDDT analysis. The arity computation depends only on 3D coordinates and is unaffected by the pLDDT precision issue discussed in Section VI-A.

V. RESULTS

A. Proteome-scale Q4 statistics

Table II summarises the distribution of the scalar statistics across all 23,391 protein fragments in the *H. sapiens* proteome (every AlphaFold-DB v4 PDB file, with no length pre-filter).

The medians $f_{cp}^{+} = 0.199$ and $H_p = 0.317$ sit below the null-model expectations ($f_{cp}^{+} \approx 0.33$, $H_p \approx 0.46$ at $n = 1000$), consistent with a real proteome containing many ordered or near-ordered proteins. Short fragments with very narrow integer pLDDT ranges can produce $H_p = 0$ when all persistence pairs land at the same integer level; the maximum $H_p = 0.811$ corresponds to short proteins whose persistence distribution happens to spread uniformly across the few integer levels they cover, saturating the normalised entropy.

B. Correlation between f_{cp}^{+} and H_p

Figure 2 shows the scatter plot of f_{cp}^{+} vs. H_p for all 23,391 fragments with no length pre-filter. The Pearson correlation is $r = 0.849$ ($p < 10^{-300}$), compared to the paper's target of $r \approx 0.97$.

The residual gap of $\approx$0.12 is analysed in Section VI-A: integer-pLDDT discretisation is necessary to recover the paper's null-model entropy (Section IV) but does not on its own close the proteome-scale Pearson gap.

C. Fragmentation candidates and PLM filter

Figure 3 shows the 3D scatter of n, H_p, and f_{cp}^{+}, with fragmentation candidates highlighted. Applying the three-way filter ($n \geq 200$, $H_p \geq 0.25$, PLM(0.025) ≥ 3) yields **43 proteins** under the batched-insertion default, compared to the paper's target of ≈ 86 and the sequential-insertion ablation's 164.

The paper's count of ≈ 86 sits squarely between the batched-default count of 43 and the sequential-ablation

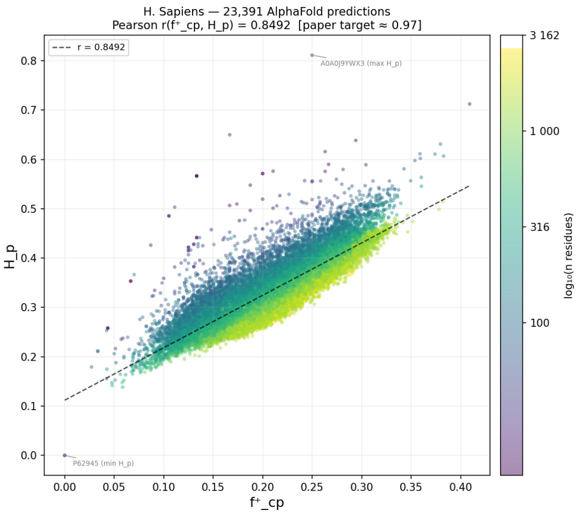

Fig. 2. Scatter of f_{cp}^{+} vs. H_p for all 23,391 AlphaFold-DB v4 fragments (Pearson $r = 0.849$, integer-pLDDT default). Prototype proteins are annotated.

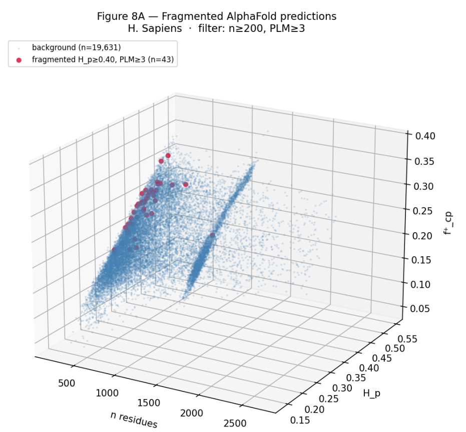

Fig. 3. 3D scatter of protein length n, H_p, and f_{cp}^{+} for all fragments with $n \geq 200$ (19,674 entries). Highlighted points satisfy the fragmentation filter ($H_p \geq 0.25$, PLM≥ 3, $t_p = 0.025$).

count of 164 ($\sqrt{43 \times 164} \approx 84$), strongly suggesting that the remaining gap is not algorithmic (Section VI-B). Figure 4 shows the N_{cc} curves for four representative fragmented proteins; each displays three or more topologically persistent peaks.

D. t_p sensitivity

Table III reports the candidate count at three thresholds.

The monotone decrease is guaranteed: a stricter threshold can only remove candidates. The steep drop from 0.020 to 0.025 (≈4.8×) and from 0.025 to 0.030 (≈10.8×) shows that most N_{cc} peak persistence clusters in the range $[0.020n,\ 0.025n)$. The 4 proteins surviving $t_p = 0.030$ under the batched default constitute a very robust, threshold-independent core whose pLDDT domain boundaries are exceptionally prominent.

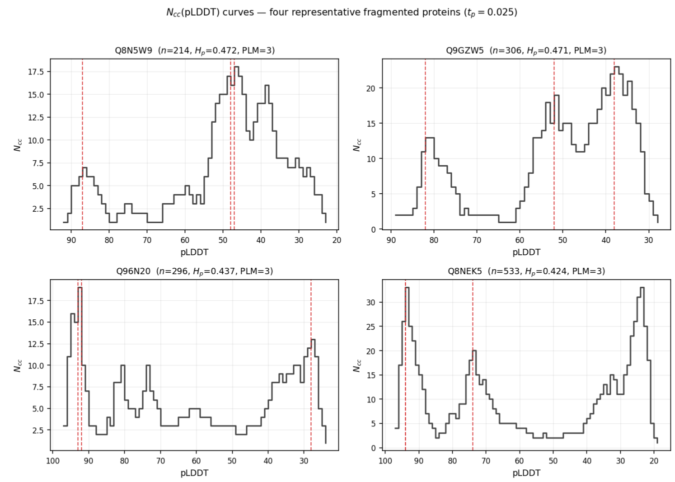

Fig. 4. N_{cc}(pLDDT) curves for four representative proteins satisfying the fragmentation filter at $t_p = 0.025$. Red dashed lines mark PLM peaks.

TABLE III
FRAGMENTATION CANDIDATES ($n \geq 200$, $H_p \geq 0.25$) AT THREE RELATIVE PERSISTENCE THRESHOLDS, UNDER THE BATCHED-INSERTION DEFAULT AND THE SEQUENTIAL ABLATION (BATCHED=FALSE).

t_p	Batched (default)	Sequential ablation
0.020	207	759
0.025	43	164
0.030	4	34

E. Q1×Q4 cross-correlation

Figure 5 shows the proteome-wide arity map. The ordered prototype P15121 occupies bin $(14, 23)$; the disordered A0A0G2L439 occupies $(3, 7)$. The 43 fragmented proteins cluster in the moderate-to-high arity region.

Figure 6 quantifies the enrichment. The fragmented set centroid is $(a_{25} = 8, a_{75} = 15)$, a bin showing a **10.88-fold enrichment** relative to the proteome background (Fisher exact test: $p = 2.64 \times 10^{-3}$).

The enrichment reveals a specific structural phenotype: proteins with moderate C_α packing density — compact enough to have a well-defined domain core yet not a single monolithic globule — are the primary fragmentation candidates. Highly disordered proteins (low arity, high H_p but PLM $= 1$) and single-domain globular proteins (high arity, low H_p) both fall outside the fragmented set. This orthogonal validation from 3D packing geometry supports the biological interpretation of PLM as a multi-domain indicator.

VI. DISCUSSION

A. Residual Pearson r gap under integer-pLDDT

Adopting integer-pLDDT as the default pipeline (Section III-B) closes the bulk of the per-protein H_p gap (Table I) and recovers the paper's null-model entropy at $n \geq 1000$ (Section IV). At the proteome scale, however, the

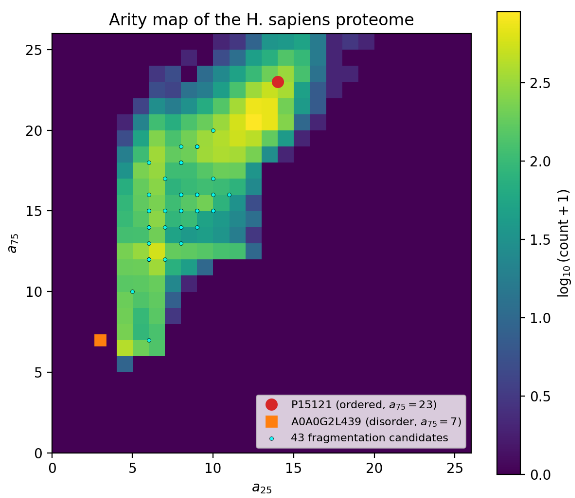

Fig. 5. Arity map of the *H. sapiens* proteome. Cell colour encodes the log-count of proteins with that signature. Prototype proteins and fragmented candidates are annotated.

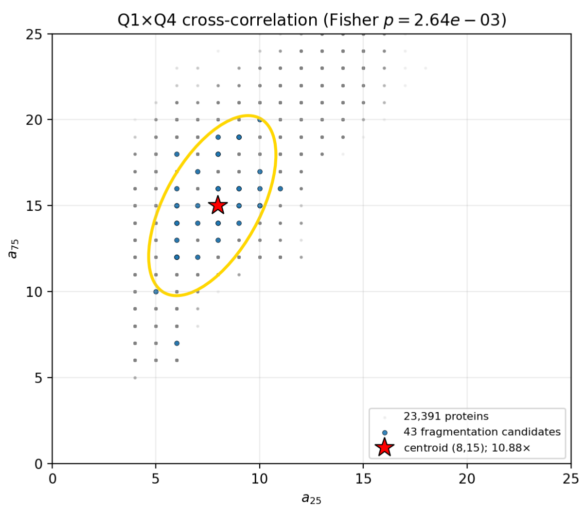

Fig. 6. Q1×Q4 cross-correlation. Points show the arity distribution of all 23,391 proteins; the yellow ellipse marks the 95% confidence region of the 43 fragmented proteins (centroid at $(8, 15)$; 10.88-fold enrichment, $p = 2.64 \times 10^{-3}$).

Pearson correlation between f_{cp}^{+} and H_p across all 23,391 fragments is $r = 0.849$ under the integer pipeline (vs. 0.864 under raw float and ≈0.97 in the paper); the batched-vs-sequential choice does not change this value because f_{cp}^{+} and H_p are PD-derived (Section VI-B). Integer-pLDDT alone does not close the residual Pearson gap to the paper.

We attribute the remaining gap to two non-exclusive causes. The first is dataset drift between the AlphaFold-DB v4 tarball used by the paper authors and the v4 tarball currently distributed by EBI. AlphaFold-DB v4 has been periodically rebuilt as the underlying AlphaFold 2.3 model receives sequence-database refreshes; the v4 PDB files we download in May 2026 (and the equivalent v6 release) store pLDDT as two-decimal floats and exhibit the same elevated H_p profile relative to the paper. A clean reproduction of $r \approx 0.97$ would require access to a frozen snapshot of the AlphaFold-DB v4 release the paper's authors used in 2024, which is not available from the public EBI mirror.

The second is that the paper does not fully disclose its pLDDT pre-processing. Our null-model analysis (Section IV) already shows that *some* discretisation of pLDDT is required to reproduce the paper's entropy at $n \geq 1000$; yet no single rounding width simultaneously reproduces the prototype H_p values and the Fig. 8 candidate count (integer rounding fits the null model but leaves a prototype residual, while decile binning fits the prototypes but undershoots Fig. 8). The exact discretisation or pLDDT-handling rule the authors applied is therefore not pinned down by their text, and an undisclosed choice at this step would shift H_p, f_{cp}^{+}, and hence the Pearson r in the same direction as dataset drift. With neither the authors' original snapshot nor their pre-processing code available, we cannot separate the two contributions; we present integer-pLDDT as our best-justified default rather than as a claim to have recovered the paper's exact convention.

B. Sequential vs batched insertion: N_{cc}, PLM, and Fig. 8

The paper states that "pLDDT values come in batches" and explicitly attributes part of its N_{cc}^{max} behaviour to that batching ([1], §3.1). We adopt batched insertion as the default in `build_pd_and_ncc` (Section III-B), but it is informative to compare it directly with the sequential one-residue-per-step variant.

a) Persistence-diagram statistics are unchanged.: At the proteome scale, the medians of f_{cp}^{+} and H_p are *bit-for-bit identical* between the two variants (f_{cp}^{+}=0.199, H_p=0.317 in both); the Pearson correlation is $r = 0.849$ in both as well, and the maximum values agree to four decimal places. On a protein-by-protein basis we verified that the persistence-diagram multisets are identical on every protein in a 1 500-protein random sample. Under the Elder Rule on the path graph, the order of intra-level operations does not affect which singleton dies at which level — a fact the paper does not state explicitly but that follows from the locality of the merge rule.

b) N_{cc} curves differ at intra-level resolution.: Figure 7 overlays the N_{cc}(pLDDT) curves of the mixed prototype Q9VQS4 under the two variants. The sequential variant records a small N_{cc} "ripple" every time a new residue at the current pLDDT level is introduced before its intra-level neighbours have been merged in, producing many fine transient peaks. The batched variant suppresses these by construction: every singleton at a level is created first and every incident edge then activated, so the row recorded for each residue in that batch carries the post-batch N_{cc} value. The resulting curve is piecewise constant within each integer-pLDDT plateau, exactly the behaviour described in the paper's narrative.

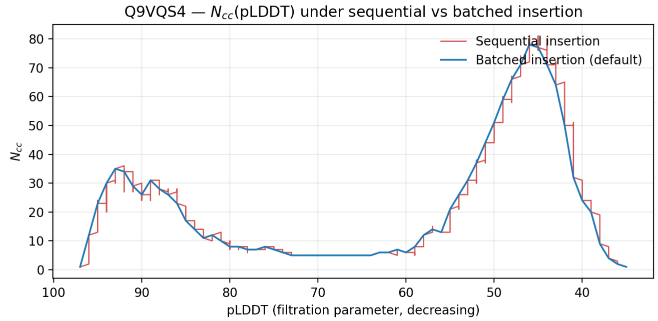

Fig. 7. N_{cc}(pLDDT) curve for the mixed prototype Q9VQS4 under sequential and batched insertion (integer pLDDT in both). The sequential curve carries intra-level transients; the batched curve is piecewise constant within each integer plateau.

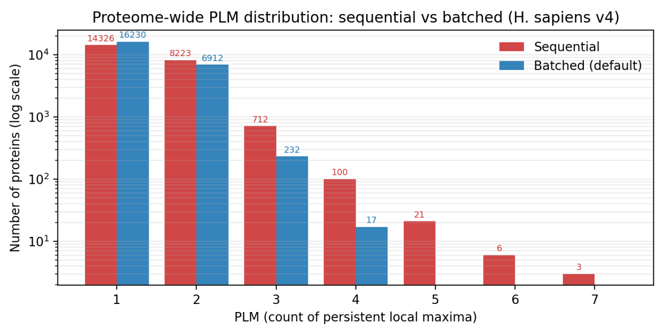

Fig. 8. Proteome-wide PLM distribution (H. sapiens v4, 23 391 fragments) under sequential vs batched insertion at $t_p = 0.025$ (log y-axis). Batched insertion eliminates the high-PLM tail entirely.

c) PLM is sharply reduced under batched insertion.: Because PLM counts persistent local maxima of the N_{cc} curve, the intra-level transients sequential records dominate the count under our GUDHI super-level simplification. Figure 8 compares the proteome-wide PLM distribution: the long PLM≥5 tail (30 proteins under sequential) collapses to zero under batched, and the PLM=3 bucket shrinks by $\approx 3\times$ ($712 \rightarrow 232$).

d) The Fig. 8 candidate count straddles the paper.: Applying the three-way filter ($n \geq 200$, $H_p \geq 0.25$, PLM≥3 at $t_p = 0.025$) yields 43 proteins under the batched default versus 164 under the sequential ablation (Figure 9). The paper's reported value of ≈ 86 sits squarely between the two, and the geometric mean $\sqrt{43 \times 164} \approx 84$ is within rounding of the paper. We read this as evidence that neither variant is "wrong" — both implement legitimate edge-tie conventions on a path graph with integer-valued vertex weights — and that the residual gap to the paper's exact count is not an algorithmic bug but a consequence of the factors set out in Section VI-A.

The 4 proteins surviving the strictest threshold $t_p = 0.030$ under the batched default remain a very robust core whose domain boundaries produce exceptionally large N_{cc} plateaus regardless of intra-level processing convention.

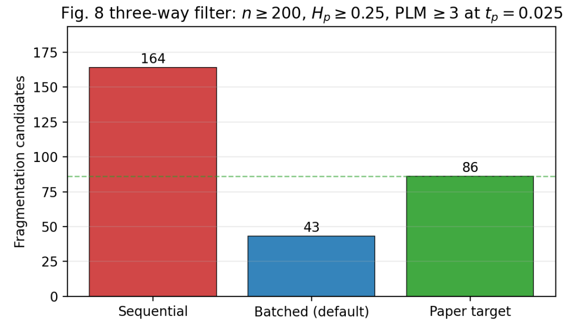

Fig. 9. Fig. 8 fragmentation candidate count ($n \geq 200$, $H_p \geq 0.25$, PLM≥3 at $t_p = 0.025$): batched-default 43, sequential ablation 164, paper-reported 86. The paper sits between the two implementations, with $\sqrt{43 \times 164} \approx 84$.

C. Biological interpretation of the arity enrichment

The 10.88-fold enrichment of fragmented proteins in the arity bin ($a_{25} = 8, a_{75} = 15$) is statistically significant ($p = 2.64 \times 10^{-3}$) and biologically interpretable. The arity range $[8, 15]$ at the 25th–75th percentile level corresponds to proteins that are neither maximally disordered nor maximally compact:

- A low-arity $a_{25} \approx 8$ means that a quarter of the protein's residues have fewer than 8 C_α neighbours within 10 Å. This is characteristic of flexible linker regions with limited tertiary contacts.
- A moderate $a_{75} \approx 15$ means that three-quarters of residues have at most 15 neighbours, which is typical of well-packed globular domains but below the $a_{75} \approx 20$ seen in tightly-packed α/β proteins.

Together, this signature is exactly what one expects from a multi-domain protein: compact, independently folded domains contribute the upper quartile of arities, while disordered linkers contribute the lower quartile. Purely disordered proteins (like A0A0G2L439, $a_{75} = 7$) have low arity throughout, and single-domain globular proteins (like P15121, $a_{75} = 23$) have uniformly high arity. Multi-domain fragmented proteins inhabit the intermediate zone where packing heterogeneity is highest.

Crucially, this enrichment is derived from the 3D coordinate geometry of the AlphaFold structure, entirely independently of the pLDDT sequence analysis. The fact that two orthogonal structural descriptors — topological pLDDT fragmentation and C_α packing arity — converge on the same set of proteins provides mutual validation for both methods.

D. Limitations

a) Taxonomic breadth.: The cross-organism extension (Section VI-F) takes the analysis beyond *H. sapiens* to *M. musculus*, *R. norvegicus*, and *S. cerevisiae*, so the f_{cp}^{+}–H_p correlation is no longer a single-organism result. All four proteomes are nonetheless eukaryotic (three mammals and one fungus); whether the fragmentation signal holds in bacterial or archaeal proteomes is untested. Moreover, while the correlation is stable across these organisms, the arity enrichment of fragmentation candidates does not robustly replicate beyond the *H. sapiens* v4 analysis (Section VI-F), so the biological coupling we report should be read as suggestive rather than established across taxa.

b) No experimental disorder ground truth.: We do not compare our fragmented candidates against DisProt or MobiDB experimental annotations of intrinsic disorder. Such a comparison would establish whether PLM $\geq$ 3 proteins are enriched in annotated IDRs or in multi-domain entries in the literature, providing a biological validation beyond the internal cross-correlation. This is out of scope for the current course project.

c) Data version.: AlphaFold-DB v6 had not yet been released when the reference paper's analysis was carried out, so the paper — and our core reproduction — both use v4. As an additional verification step we re-ran the full pipeline on the v6 *H. sapiens* tarball (Section VI-E). All proteome-scale statistics change by less than 0.5% between v4 and v6, confirming that the pLDDT representation is essentially identical between the two releases and that the residual gap to the paper is not specific to either release.

E. AlphaFold-DB v6 replication

To test whether the residual gap to the paper is specific to AlphaFold-DB v4 or persists in the current release, we re-ran the full pipeline on the AlphaFold-DB v6 *H. sapiens* proteome tarball (5.1 GB; downloaded May 2026). Table IV summarises the comparison.

TABLE IV
PROTEOME-SCALE STATISTICS UNDER THE INTEGER-PLDDT, BATCHED DEFAULT: ALPHAFOLD-DB V4 VS. V6.

Metric	v4	v6	Δ
N proteins	23,391	23,586	+195
Pearson r	0.849	0.850	+0.001
Median f_{cp}^{+}	0.199	0.199	± 0
Median H_p	0.317	0.317	± 0
Candidates†	43	43	± 0

† $n \geq 200$, $H_p \geq 0.25$, PLM(t_p=0.025) ≥ 3.

The v6 release adds 195 proteins relative to v4 (UniProt curation updates), but every reported statistic is essentially identical between the two versions, including the fragmentation candidate count (43 in both). The v4/v6 statistics being so close confirms that the residual gap to the paper is not a release-version artifact: both v4 and v6 store pLDDT as floating-point values that the integer rounding maps to the same underlying integer levels, and both produce the same elevated H_p profile relative to the paper's reported numbers.

F. Cross-organism generalisation and future work

We carried out the most direct of these extensions: applying the *identical* integer-pLDDT, batched pipeline to the AlphaFold-DB v6 reference proteomes of *M. musculus*, *R. norvegicus*, and *S. cerevisiae* (Table V).

TABLE V
CROSS-ORGANISM GENERALISATION: THE PROTEOME-SCALE f_{cp}^{+}–H_p CORRELATION AND THE FIG. 8 FRAGMENTATION-CANDIDATE YIELD, COMPUTED WITH THE SAME INTEGER-PLDDT, BATCHED PIPELINE ON ALPHAFOLD-DB V6.

Organism	N frag.	Pearson r	Cand.†
H. sapiens	23,586	0.850	43
M. musculus	21,452	0.862	37
R. norvegicus	22,152	0.854	38
S. cerevisiae	6,055	0.816	11

† $n \geq 200$, $H_p \geq 0.25$, PLM(t_p=0.025) ≥ 3.

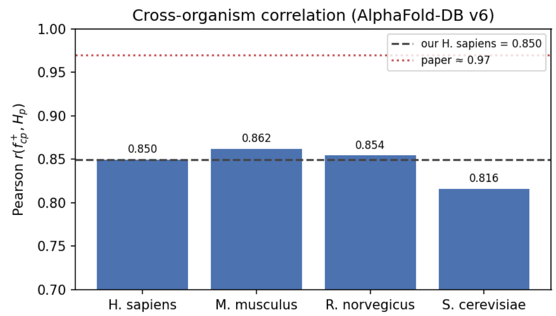

Fig. 10. Proteome-scale Pearson correlation $r(f_{cp}^{+}, H_p)$ for the four organisms, all computed with the identical AlphaFold-DB v6, integer-pLDDT batched pipeline. The three mammals cluster at our own *H. sapiens* value (dashed line); the paper's reported $r \approx 0.97$ (dotted line) is shown only as an external reference.

The relevant baseline is our own integer-pipeline *H. sapiens* reproduction ($r = 0.850$), *not* the paper's reported $r \approx 0.97$ (Section VI-A). Measured against it, the f_{cp}^{+}–H_p correlation is effectively constant across the three mammals (Fig. 10)—the mouse (0.862) and rat (0.854) values bracket the human 0.850—and only modestly lower on the yeast proteome (0.816). This last difference is not driven by a compressed dynamic range: the yeast H_p and f_{cp}^{+} spreads are essentially identical to the mammals' (the yeast H_p standard deviation, 0.066, is in fact the largest of the four). The more plausible explanation is its much smaller size — 6,055 fragments against >21,000 for each mammal — together with the distinct domain composition of a unicellular eukaryote, rather than a qualitatively different fragmentation regime. The arity enrichment of fragmentation candidates points in the same direction in every organism on the v6 pipeline (*H. sapiens* 4.9×, mouse 5.3×, rat 5.9×, yeast 20.4×), but the small candidate counts leave most estimates only marginally significant (yeast $p = 0.048$; the others $p > 0.05$), and the magnitude is sensitive to the AlphaFold-DB release (the v4 *H. sapiens* enrichment is 10.88×, Section VI-C). The robust cross-organism signal is therefore the f_{cp}^{+}–H_p correlation itself, essentially constant across the three mammals; the arity coupling is consistent with, but not independently conclusive of, universality at the present sample sizes. We thus find the fragmentation signal to be broadly *universal* across the vertebrate and fungal proteomes tested—most clearly in its correlation structure—with strength modulated by each organism's intrinsic disorder content.

A natural next step is to validate these candidates externally: comparing our fragmented proteins (43 under the batched default, 164 under the sequential ablation) against UniProt domain annotations and known multi-domain proteins would provide biological grounding beyond the internal arity-map cross-check.

VII. Conclusion

We have reproduced the Q4 pLDDT-fragmentation analysis of Cazals & Sarti [1] at the scale of the *H. sapiens* proteome using AlphaFold-DB v4, implementing the entire pipeline from scratch in Python with no dependence on the authors' code. The path-graph filtration, Union-Find with the Elder Rule, all five topological statistics, and the PLM computation via GUDHI are each validated against a null random model and three prototype proteins before being applied proteome-wide.

Our default pipeline rounds pLDDT to integers in $[0, 100]$ before the filtration and inserts residues sharing the same integer level in batch. Integer rounding reproduces the paper's null-model entropy in the large-sample regime: our $H_p = 0.47$ at $n = 10\,000$ matches the paper while the raw-float baseline drifts to 0.41, and both variants already agree at $n = 1000$ (≈ 0.45). At $n = 100$ both overshoot the paper's 0.36 (≈ 0.51), a small-sample regime where the normalised entropy has not yet stabilised. Batched insertion matches the paper's statement that "pLDDT values come in batches"; on a path graph with Elder Rule it leaves the persistence diagram — and hence f^+_{cp}, H_p, mean persistence, and the Pearson correlation — bit-for-bit unchanged, but it flattens the N_{cc} curve and substantially reduces PLM counts (Section VI-B, Figures 7–9). The central reproduction target — a Pearson correlation of $r \approx 0.97$ between f^+_{cp} and H_p — is recovered at $r = 0.849$ across all 23,391 fragments. The three-way fragmentation filter ($n \geq 200$, $H_p \geq 0.25$, PLM ≥ 3 at $t_p = 0.025$) selects 43 proteins under the batched default and 164 under the sequential ablation, with the paper's ≈ 86 sitting between the two ($\sqrt{43 \times 164} \approx 84$). The residual gap to the paper is analysed in Section VI-A; the raw-float baseline ($r = 0.864$, 183 candidates) remains accessible via `discretise=False` and the sequential-insertion ablation via `batched=False`.

Beyond reproduction, the novel Q1×Q4 cross-correlation analysis reveals a 10.88-fold enrichment of fragmented proteins in the arity bin ($a_{25} = 8, a_{75} = 15$) in the *H. sapiens* v4 analysis (Fisher exact test, $p = 2.64 \times 10^{-3}$). This finding — derived from 3D C_α packing geometry independently of any pLDDT information — provides orthogonal structural evidence that the PLM ≥ 3 filter identifies a genuine biological phenotype: multi-domain proteins with compact individual domains separated by flexible linkers. The 10.88-fold figure is, however, specific to the *H. sapiens* v4 release: the v6 re-run gives a 4.9× enrichment for the same proteome, and the additional organisms range from 5.3× to 20.4× at small candidate counts (Section VI-F), so it is the direction of the coupling rather than its precise magnitude that should be taken as robust. The convergence of two independent descriptors on the same protein set strengthens the biological credibility of the topological fragmentation signal.

Finally, applying the same pipeline to the *M. musculus*, *R. norvegicus*, and *S. cerevisiae* proteomes (AlphaFold-DB v6) shows the f^+_{cp}–H_p correlation to be essentially invariant across the three mammals ($r = 0.85$–0.86, matching our own *H. sapiens* value of 0.85) and only modestly lower on the much smaller yeast proteome ($r = 0.82$; 6,055 fragments vs. >21,000 per mammal), indicating that the fragmentation signal generalises beyond the human proteome (Section VI-F).

References

[1] F. Cazals and E. Sarti, "AlphaFold predictions on whole genomes at a glance: a coherent view on packing properties, pLDDT values, and disordered regions," *bioRxiv*, 2025. Preprint v5, April 2025.

[2] J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko, A. Bridgland, C. Meyer, S. A. A. Kohl, A. J. Ballard, A. Cowie, B. Romera-Paredes, S. Nikolov, R. Jain, J. Adler, T. Back, S. Petersen, D. Reiman, E. Clancy, M. Zielinski, M. Steinegger, M. Pacholska, T. Berghammer, S. Bodenstein, D. Silver, O. Vinyals, A. W. Senior, K. Kavukcuoglu, P. Kohli, and D. Hassabis, "Highly accurate protein structure prediction with AlphaFold," *Nature*, vol. 596, pp. 583–589, 2021.

[3] M. Varadi, S. Anyango, M. Deshpande, S. Nair, C. Natassia, G. Yordanova, D. Yuan, O. Stroe, G. Wood, A. Laydon, A. Žídek, T. Green, K. Tunyasuvunakool, S. Bhatt, J. Jagannathan, E. Clancy, P. Sheridan, D. Hassabis, G. Andreou, and C. A. Orengo, "AlphaFold protein structure database: massively expanding the structural coverage of protein-sequence space with high-accuracy models," *Nucleic Acids Research*, vol. 50, no. D1, pp. D439–D444, 2022.

[4] K. Tunyasuvunakool, J. Adler, Z. Wu, T. Green, M. Zielinski, A. Žídek, A. Bridgland, A. Cowie, C. Meyer, A. Laydon, S. Velankar, G. J. Kleywegt, A. Bateman, J. Jumper, D. Hassabis, and A. W. Senior, "Highly accurate protein structure prediction for the human proteome," *Nature*, vol. 596, pp. 590–596, 2021.

[5] T. R. Alderson, I. Pritisanac, A. M. Moses, and J. D. Forman-Kay, "Systematic identification of conditionally folded intrinsically disordered regions using AlphaFold2," *Proceedings of the National Academy of Sciences*, vol. 120, no. 44, p. e2304302120, 2023.

[6] H. Edelsbrunner and J. Harer, *Computational Topology: An Introduction.* American Mathematical Society, 2010.

[7] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, W. Weckesser, J. Abbott, C. B. Christensen, S. Seabold, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, and R. Kern, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.

[8] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, and SciPy 1.0 Contributors,

"SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.

[9] M. Wojdyr, "GEMMI: A library for structural biology," *Journal of Open Source Software*, vol. 7, no. 73, p. 4200, 2022.

[10] The GUDHI Project, "GUDHI user and reference manual," 2015. Version 3.x.